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Not Everyone Ages Gracefully: Fiscal Adjustments to Exogenous Shocks in Three Latin American Countries

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ABSTRACT

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1. Introduction

- Motivation
  - Global challenges in fiscal sustainability.
  - Importance on focusiig on latin-american economies
  - Unique economic and demographic profiles of Mexico, Chile, and Costa Rica.
  - Importance of worker heterogeneity in fiscal adjustments.
- Research Questions
- How should public expenditure be adjusted to minimize fiscal imbalances considering different skill groups?
- What replacement rate minimizes the fiscal gap without altering contribution rates?
- How do pension reforms affect different worker types (high vs. low skill)?
- What are the policy trade-offs for Latin American countries with different fiscal constraints?

2. Model

2.1. Introduction

A Dynamic Stochastic Heterogeneous Agent Overlapping Generations (DSOLG) Model was developed to estimate changes in fiscal policy.

Unlike infinite-horizon representative-agent models, this approach allows for the incorporation of [1] :

- Life cycle characteristics that are important for determining savings and labor supply choices.
- Intra-generational heterogeneity in households, which is necessary to analyze the impact of policy changes on income and wealth distribution.
- Inter-generational heterogeneity in households to analyze the timing of taxes and their effects on intergenerational distribution.

One generation of OLG models are those that incorporate uncertainty in form of idiosyncratic shocks at the household level (labor income, longevity risk, etc.) and determinism in aggregate variables [1]

Idiosyncratic shocks affect agents in different ways, respond heterogeneously within a cohort [2][1]

These models are used to calculate transition effects of policy changes from one steady state to the next.

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[1] In contrast to the representative agent approach where it is implicitly defined that individuals can protect themselves against any form of idiosyncratic shock

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The dynamics of the transition are also used to analyze the welfare impacts of fiscal policy reforms that may benefit future generations at the expense of the transition generations.

The model presented here belongs to this generation of OLG models. It is a dynamic and stochastic general equilibrium model that incorporates idiosyncratic risks in labor productivity. The aggregate quantities of the economy grow on a balanced growth path given by the population growth rate. n_p .

2.2. Demography

In each period t , The economy is populated by J overlapping generations indexed by $j = 1, ..., J$.

The model integrates the so-called **intensive margin of informality**, specifically workers who are employed in formal economic units but who do not have an employer relationship, labor benefits defined by law, or social security.[2] When individuals enter the labor market, they are assigned as informal or formal workers according to a probability distribution ω_s[3] . The indicator variable $m_s \in [0,1]$ denotes the worker's employment status, where $m_s = 0$ This applies to formal workers and $m_s = 1$ to informal workers. The probability of transition between both states is fixed and does not depend on age:

$$\pi_{j,m,m^+} = \Pr(m_{j+1} = m^+ \mid m_j = m) \quad \text{con} \quad m, m^+ \in \{0,1\}, \tag{1}$$

It is assumed that survival from one period to the next is stochastic and that ψ_j is the probability that an agent will survive to old age $j-1$ at the age j , conditional on living at the age $j-1$[4].

The unconditional probability of surviving to old age j is given by $\Pi_{i=1}^{j}\psi_i$ with $\psi_1 = 1$. Since the number of members in each cohort declines with age, the cohort size corresponding to age j in the period t is

$$N_{j,s,t} = \psi_{j,t} N_{j-1,s,t-1} \quad \text{con} \quad N_{1,s,t} = (1 + n_{p,t}) N_{1,s,t-1} \tag{2}$$

Consequently, cohort weights (relative population ratios) are defined as $m_{1,s,t} = 1$ y $m_{j,s,t} = \frac{\psi_{j,t}}{1+n_{p,t}} m_{j-1,s,t-1}$.

It is assumed that the population grows at a constant rate. $n_{p,t} = n_p$ and it is the same for both population groups.

The balanced growth trajectory, where all aggregate variables grow at the same rate, is set to the growth rate of the youngest cohort. These aggregate variables are then normalized over time t due to the size of the younger cohort living during that period.

All officers retire at age j_r . Agents who worked in the formal sector begin receiving a pension, which is funded by payroll tax. During their working years, formal sector employees accumulate earning points ep_j that define their pension payments when they retire.

For simplicity, we will omit the index s as much as possible.

2.3. Household decisions

2.3.1. Phousehold references

Individuals have consumption preferences $c_{j,t}$ and leisure $l_{j,t}$, They also pay consumption and income taxes, as well as a payroll tax to the pension system. It is assumed that the time allocation is equal to 1.

With $l_{j,t}$ denoting the amount of labor hours offered to the market in the period t , we have $\mathrm{l}_{j,t} + l_{j,t} = 1$. The utility function of households is defined as

$$E\left[\sum_{j=1}^{J} \beta^{j-1} \left(\Pi_{i=2}^{j} \psi_{i,s} m_{i-1,s}\right) u(c_{j,s}, 1 - l_{j,s})\right] \tag{3}$$

where β denotes the time discount factor. As can be seen, the expected marginal utility of future consumption is also conditional on the current employment status m .

The household utility function is given by

$$u(c_{j,t}, 1 - l_{j,t}) = \frac{\left[(c_{j,t})^{\nu}(1 - l_{j,t})^{(1-\nu)}\right]^{\left(1-\frac{1}{\gamma}\right)}}{1 - \frac{1}{\gamma}} \tag{4}$$

The consumption and leisure utility takes the form of a Cobb-Douglas function with one parameter ν preference between leisure and consumption. The intertemporal elasticity of substitution is constant and equal to γ , where $\frac{1}{\gamma}$ It is the household's risk aversion.

2.3.2. Risk of survival and inheritance

Since there are no annuity markets, the return on individual assets corresponds to the net interest rate.

In a scenario where there is no risk of longevity, individuals know with certainty when their lives will end. Consequently, they are able to perfectly plan when they want to spend all their savings.

[2]The **intensive margin of informality** is even larger, as it includes workers employed in the **informal sector**. INEGI defines the informal sector as economic activities that operate with household resources, without being formally constituted as businesses, where it is impossible to distinguish between the economic unit and the household. In other words, there are two ways to conceptualize informality: according to the economic sector where the worker is employed and according to their employment status. The model incorporates **informal** workers employed in formal economic units and **formal** workers. The model could consider informal workers employed in the informal sector by considering economic units that face a production function that uses only the labor factor.

[3]This distribution is calculated empirically using the INEGI Hussmans matrix

[4]It is assumed that formal and informal workers have the same survival rate

There is uncertainty about survival here, so agents may die before reaching their maximum lifespan J and, as a consequence, leave an inheritance. It is defined $b_{j,t}$ like the inheritance that an agent at age j receives in the period t .

The amount of inheritance for each cohort can be calculated using the expression:

$$b_{j,t} = \Gamma_{j,t} BQ_t \tag{5}$$

where BQ_t defines the aggregate inheritance in the period t , or simply the fraction of total assets that can be attributed to those who died at the end of the previous period (including interest).

$$BQ_t = r_t^n \sum_{j=2}^{J} a_{j,t} \frac{m_{j,t}}{\psi_{j,t}} (1 - \psi_{j,t}) \tag{6}$$

where r_t^n is the net interest rate in t and $a_{j,s,t}$ are the cohort's assets j , of the group s , in t .

2.3.3. Risk to labor productivity

Individuals differ with respect to their work productivity $h_{j,t}$, which depends on a (deterministic) income profile by age $e_{j,s}$ which depends on the type of work, a fixed productivity effect θ which is defined at the beginning of the life cycle and which, likewise, depends on the type of work (formal and informal) to which they are assigned[5]. In addition, an idiosyncratic shock is added through an autoregressive component $\eta_{j,t}$ that evolves over time and has an autoregressive structure of order 1, so that

$$\eta_j = \rho \eta_{j-1} + \epsilon_j \quad \text{con} \quad \epsilon_j \sim N(0, \sigma_\epsilon^2) \quad \text{y} \quad \eta_1 = 0 \tag{7}$$

Given this structure, household labor productivity is

$$h_j = \begin{cases} e_j \exp[\eta_j] & \text{si } j < j_r \\ 0 & \text{si } j \geq j_r \end{cases} \tag{8}$$

2.3.4. Consumer Decision Problem

The state of individuals is characterized by the state vector[6]

$$z_j = (J, a, ep, s, \eta) \tag{9}$$

Households maximize the utility function subject to the intertemporal budget constraint

$$a_{j+1,s,t} = \begin{cases} (1+r_t^n) a_{j,s,t-1} + w_t^n h_{j,s,t} l_{j,s,t} + b_{j,s,t} + pen_{j,s,t} - p_t c_{j,s,t} & \text{si } s = 0 \\ (1+r_t^n) a_{j,s,t-1} + w_t h_{j,s,t} l_{j,s,t} + b_{j,s,t} - p_t c_{j,s,t} & \text{si } s = 1 \end{cases} \tag{10}$$

where:

- $a_{j,t}$ These are the agent's savings assets in the period t ,
- $w_t^n = w_t\left(1 - \tau_t^w - \tau_{j,t}^{impl}\right)$ It is the net wage rate, which is equal to the market wage w_t less taxes on employment income $\tau_{j,t}^{impl}$ and the payroll tax to finance the pension system τ_t^p
- $r_t^n = r_t(1 - \tau_t^r)$ It is the net interest rate, which is equal to the market interest rate. r_t deducting the capital gains tax τ_t^r ,
- $p_t = 1 + \tau_t^c$ It is the consumer price which is normalized to one and the consumption taxes are added τ_t^c .

An additional non-negativity restriction on savings is added. $a_{j+1,s} \geq 0$

2.4. Dynamic programming problem

The agent optimization problem is as follows:

$$\begin{aligned} V_t(j, a, ep, s, \eta) &= \max_{c,l,a^+,ep^+} u(c, 1-l) + \beta \psi_{j+1}(m_s) E[V(j+1, a^+, ep^+, s^+, \eta^+) \mid \eta, m_s] \\ \text{s.t. } a^+ &= \begin{cases} (1+r_t^n) a_{j,s,t-1} + w_t^n h_{j,s,t} l_{j,s,t} + b_{j,s,t} + pen_{j,s,t} - p_t c_{j,s,t} & \text{si } s = 0 \\ (1+r_t^n) a_{j,s,t-1} + w_t h_{j,s,t} l_{j,s,t} + b_{j,s,t} - p_t c_{j,s,t} & \text{si } s = 1 \end{cases}, \\ \eta^+ &= \rho \eta + \epsilon^+ \quad \text{con} \quad \epsilon^+ \sim N(0, \sigma_\epsilon^2) \\ \pi_{j,m,m^+} &= \Pr\left(m_{j+1,s} = m_s^+ \mid m_{j,s} = m_s\right) \quad \text{con} \quad m_s, m_s^+ \in \{0, 1\}. \end{aligned} \tag{11}$$

where $z = (j, a, ep, s, \eta)$ is the vector of individual state variables. Note that a time index was placed on the value function and on the prices. This is necessary to calculate the dynamics of the transition between two stationary states. The terminal condition of the value function is

$$V_t(z) = 0 \quad \text{para} \quad z = (J+1, a, ep, s, \eta) \tag{12}$$

which means that it is assumed that the agents do not value what happens after death.

We formulate the solution to the household problem by recognizing that we can write the working hours and consumption functions as functions of a^+ :

[5] It represents a permanent shock

[6] It is assumed that productivity shocks are independent between individuals and identically distributed among individuals of a specific type of work.

$$
\begin{aligned}
l = l(a^+) &= \min\left\{\max\left[\nu + \frac{1-\nu}{(w_t^n*(1-m)+w_t*m)h}(a^+ - (1+r_t^n)a - \text{pen }*(1-m)),0\right],1\right\} \\
c = c(a^+) &= \frac{1}{p_t}[(1+r_t^n)a + (w_t^n*(1-m)+w_t*m)hl(a^+) + \text{pen }*(1-m) - a^+]
\end{aligned}
\tag{13}
$$

With the definition of the **implicit tax rate** (See next section), The first-order conditions of households are defined as

$$
\begin{aligned}
&\frac{v}{p_t}\cdot\frac{\left[c^\nu(1-l)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c} = \beta E[V_{a^+}(z^+)\mid\eta] \\
&\frac{1-\nu}{v}\cdot p_t c = w_t h(1-l)\left\{1-\tau_t^w-\tau_{j,t}^{impl}\right\}
\end{aligned}
\tag{14}
$$

where a^+ is unknown. Note that $\tau_{j,t}^{impl} = \tau_t^p$ para $\lambda = 1$, which reduces to the original model.

2.4.1. Household income and expenses through the pension system

At the mandatory retirement age j_r , labor productivity falls to zero and Households receive a pension $pen_{j,t}$, which is based on the individual's salary history. In order to track and account for past salaries as well as pension contributions, an **earning points** statement is added, ep , which captures individual gross incomes relative to the average income of the entire economy for each year of contribution[7] [3]

$$
ep_{j+1} = \left[ep_j \times (j-1) + \left(\lambda + (1-\lambda)\frac{whl_j}{\bar{y}}\right)\right]/j \tag{15}
$$

where the parameter λ indicates the level of **progressivity** of the pension system [2]. When $\lambda = 1$ The pension is independent of previous contributions and is equal to the fraction of the pension system's replacement rate κ of the average labor income in the period t , this is:

$$
\text{pen}_j = \begin{cases} 0 & \text{si } j < j_r \\ \kappa_t \frac{w_t}{j_r - 1}\sum_{j=1}^{j_r-1} e_j, & \text{si } j \geq j_r \end{cases} \tag{16}
$$

When $\lambda = 0$, The pension depends entirely on the worker's salary history. During the worker's retirement phase $j \geq j_r$, Salary points remain constant and the pension is calculated as

$$
\text{pen}_j = \kappa_t \times ep_{j_r} \times \bar{y} \tag{17}
$$

The earning points evolve according to the equation:

$$
ep^+ = \begin{cases} \frac{j-1}{j}\cdot ep + \frac{1}{j}\cdot\left[\lambda + (1-\lambda)\cdot\frac{w_t hl}{y_t}\right] & \text{si } j < j_r, \\ ep & \text{en caso contrario.} \end{cases} \tag{18}
$$

This equation integrates two parts:
- The accumulation phase, $j < j_r$
- The yield phase, $j \geq j_r$

The household budget constraint changes to:

$$
a^+ + p_t c = (1+r_t^n)a + w_t^n hl + \mathbb{1}_{j\geq j_r}\kappa_t \bar{y}_t ep \tag{19}
$$

The pension benefit is received until retirement age is reached j_r and is equal to the product of the current replacement rate κ_t , the average income $\bar{y}$ as well as the points of income accumulated by the worker ep .

The Lagrangian of the household optimization problem is written:

$$
\begin{aligned}
\mathcal{L} = &\frac{\left[c^v(1-l)^{1-v}\right]^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta E[V(z^+)\mid\eta] + \\
&+\mu_1\left[(1+r_t^n)a + w_t^n hl + \mathbb{1}_{j\geq j_r}\kappa_t\bar{y}_t ep - a^+ - p_t c\right] \\
&+\mu_2\mathbb{1}_{j<j_r}\left[\frac{j-1}{j}\cdot ep + \frac{1}{j}\cdot\left[\lambda + (1-\lambda)\cdot\frac{w_t hl}{\bar{y}_t}\right] - ep^+\right] \\
&+\mu_2\mathbb{1}_{j\geq j_r}[ep - ep^+]
\end{aligned}
\tag{20}
$$

with the following first-order conditions:

$$
\begin{aligned}
&\frac{\nu}{p_t}\cdot\frac{\left[c^\nu(1-l)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c} = \beta E[V_{a^+}(z^+)\mid\eta] \\
&\frac{1-\nu}{\nu}\cdot p_t c = w_t h(1-l)\left\{1-\tau_t^w-\tau_t^p + \frac{1-\lambda}{j\cdot\bar{y}}\cdot\frac{\beta E\left[V_{ep^+}(z^+)\mid\eta\right]}{\frac{\nu}{p_t}\cdot\frac{\left[c^\nu(1-l)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c}}\right\}
\end{aligned}
\tag{21}
$$

When $\lambda = 1$, the part of the equation

$$
\frac{1-\lambda}{j\cdot\bar{y}}\cdot\frac{\beta E\left[V_{ep^+}(z^+)\mid\eta\right]}{\frac{\nu}{p_t}\cdot\frac{\left[c^\nu(1-l)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c}} \tag{22}
$$

It becomes zero, and we are faced with the base case of a flat pension, independent of previous contributions.

[7]This revenue tracking mechanism is taken from [3], which is how the German pension system works.

From the envelope theorem we obtain:

$$\begin{aligned} V_{a^+}(z^+) &= (1+r^n_{t+1}) \cdot \frac{v}{p_{t+1}} \cdot \frac{\left[(c^+)^{\nu}(1-l^+)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c^+} \\ V_{ep^+}(z^+) &= \mathbb{1}_{j+1\geq j_r} \cdot \kappa_{t+1}\bar{y}_{t+1} \cdot \frac{\nu}{p_{t+1}} \cdot \frac{\left[(c^+)^{\nu}(1-l^+)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c^+} \\ &\quad + \left[\mathbb{1}_{j+1<j_r} \cdot \frac{j}{j+1} + \mathbb{1}_{j+1\geq j_r}\right] \cdot \beta E\left[V_{ep^{++}}(z^{++}) \mid \eta\right]. \end{aligned} \tag{23}$$

Iterating the first-order condition forward, we obtain:

$$\beta^{i-j} E\left[\frac{v}{p_{t+i-j}} \cdot \frac{\left[(c_i)^{v}(1-l_i)^{1-v}\right]^{1-\frac{1}{\gamma}}}{c_i} \mid \eta_j\right] = \frac{\frac{v}{p_t} \cdot \frac{\left[(c_j)^{v}(1-l_j)^{1-v}\right]^{1-\frac{1}{\gamma}}}{c_j}}{\prod_{k=j+1}^{i}\left(1+r_{t+k-j}\right)} (I) \tag{24}$$

For $i < j$.

For age $j+1 = j_r$, the second equation of the envelope reduces to

$$V_{ep}\left(z_{j_r}\right) = \kappa_{t+1}\bar{y}_{t+1} \cdot \frac{v}{p_{t+1}} \cdot \frac{\left[\left(c_{j_r}\right)^{\nu}\left(1-l_{j_r}\right)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c_{j_r}} + \beta E\left[V_{ep}\left(z_{j_r+1}\right) \mid \eta_j\right] (II) \tag{25}$$

with $s = t+1+i-j_r$. For any $j+1 < j_r$, we have

$$\begin{aligned} V_{ep}\left(z_{j+1}\right) &= \frac{j}{j+1} \cdot \beta E\left[V_{ep}\left(z_{j+2}\right) \mid \eta_j\right] \\ &= \frac{j}{j+1} \cdot \beta E\left[\frac{j+1}{j+2} \cdot \beta E\left[V_{ep}\left(z_{j+3}\right) \mid \eta_j\right] \mid \eta_j\right] \\ &= \frac{j}{j+2} \cdot \beta^2 E\left[V_{ep}\left(z_{j+3}\right) \mid \eta_j\right] = \ldots \\ &= \frac{j}{j_r-1} \cdot \beta^{j_r-(j+1)} E\left[V_{ep}\left(z_{j_r}\right) \mid \eta_j\right] \end{aligned} \tag{26}$$

Substituting $(\mathbb{I})$ we have

$$\beta E\left[V_{ep^+}(z^+) \mid \eta_j\right] = \frac{j}{j_r-1} \sum_{i=j_r}^{J} \kappa_s \bar{y}_s \cdot \beta^{i-j} E\left[\frac{v}{p_s} \cdot \frac{\left[(c_i)^{v}(1-l_i)^{1-v}\right]^{1-\frac{1}{\gamma}}}{c_i} \mid \eta_j\right], \tag{27}$$

with $s = t+i-j$. Now substituting the equation (I) , we have:

$$\frac{1-\lambda}{j \cdot \bar{y}_t} \cdot \frac{\beta E\left[V_{ep^+}(z^+) \mid \eta\right]}{\frac{\nu}{p_t} \cdot \frac{\left[c^{\nu}(1-l)^{1-\nu}\right]^{1-\frac{1}{\gamma}}}{c}} = \frac{1-\lambda}{(j_r-1) \cdot \bar{y}_t} \cdot \sum_{i=j_r}^{J} \frac{\kappa_s \bar{y}_s}{\prod_{k=j+1}^{i}\left(1+r_{t+k-j}\right)} \tag{28}$$

with $s = t+i-j$.

Therefore, we define the **implicit tax rate** of the pension system as

$$\tau^{impl}_{j,t} = \tau^p_t - \frac{1-\lambda}{(j_r-1) \cdot \bar{y}_t} \cdot \sum_{i=j_r}^{J} \frac{\kappa_s \bar{y}_s}{\prod_{k=j+1}^{i}\left(1+r_{t+k-j}\right)} \tag{29}$$

This implicit tax rate takes into account that, if $\lambda < 1$, Pension payments increase as employment income increases, and consequently, pension contributions also increase. Therefore, contributions τ^p_t They are different for each household.

2.5. Technology

Firms hire capital K_t and labor L_t in a perfectly competitive factor market to produce a single good Y_t according to a production technology given by a Cobb-Douglas production function

$$Y_t = \Omega K_t^{\alpha} L_t^{1-\alpha} \tag{30}$$

where Ω It is the level of technology that is constant over time. Capital depreciates at a rate δ , so the capital stock evolves according to the following expression

$$\left(1+n_p\right)K_{t+1} = (1-\delta)K_t + I_t \tag{31}$$

Under the assumption of perfect competition, the inverse functions of the firm's demand for capital and labor are given by

$$\begin{aligned} r_t &= \alpha\Omega\left[\frac{L_t}{K_t}\right]^{(1-\alpha)} - \delta \\ w_t &= (1-\alpha)\Omega\left[\frac{K_t}{L_t}\right]^{\alpha} \end{aligned} \tag{32}$$

2.6. Government

The government manages two systems: a tax system and a pension system, both operating in budgetary balance.

The government collects taxes on consumption expenditure, labor income, and capital income to finance its public spending G_t and debt-related payments B_t . In the initial equilibrium, public spending equals a constant ratio of GDP, that is, $G = g_y Y$. In subsequent periods, the level of public goods remains constant (per capita), meaning that $G_t = G$. The same applies to public debt, where the initial ratio is denoted b_y . At any point in time, the tax system's budget is balanced if the following equality holds:

$$\tau_t^c C_t + \tau_t^w w_t L_t^s + \tau_t^r r_t A_t + (1 + n_p) B_{t+1} = G_t + (1 + r_t) B_t \tag{33}$$

In addition to tax revenues, the government finances its spending by taking on new debt $(1 + n_p) B_{t+1}$. However, it must repay the existing debt, including interest on the payments, so we have to add $(1 + r_t) B_t$ to government consumption on the expenditure side. Thus, in a steady-state equilibrium, spending $(r - n_p) B$ reflects the cost needed to keep the debt level constant. Note that no a priori constraint has been placed on which tax rate needs to be adjusted to balance the budget over time.

The pension system operates on a pay-as-you-go scheme, meaning it collects contributions from the working-age generation and distributes them directly to current retirees. The pension system's budget balance equation is given by

$$\tau_t^p w_t L_t^{\text{supply}, s=0} = \text{pen}_t N^R \quad \text{con} \quad N^R = \sum_{j=j_r}^{J} m_{j,s=0} \psi_j \tag{34}$$

where N^R denotes the number of formal retirees.

It is assumed that the replacement rate κ is given exogenously while the contribution rate τ_t^p is adjusted with the aim of balancing the budget.

The pension benefit is calculated by the sum of earnings points accumulated during the working period and the *current pension amount*, APA[8] which reflects the monetary value of each earning point, multiplied by the replacement rate κ

$$p_j = \kappa \times \text{ep}_{j_r} \times \text{APA} \tag{35}$$

Over time, APA grows with gross labor income.

2.7. Market

There are three markets in the economy: the capital market, the labor market, and the goods market. With respect to the factor markets, the price of capital r_t and labor w_t adjust to clear the market; that is:

$$K_t + B_t = A \quad \text{y} \quad L_t = L_t^s \tag{36}$$

Note that there are two sectors that demand savings from households. The business sector uses savings as capital in the production process, while the government uses them as public debt to finance its spending. The government and businesses compete in perfect competition in the capital market.

With respect to the goods market, all products produced must be used either for consumption by the private sector or the government, or as investment in the future capital stock. Thus, equilibrium in the goods market is given by

$$Y_t = C_t + G_t + I_t \tag{37}$$

3. Model calibration and parametrization

The model considers two types of parameters: those that can be directly observed in the data and those that can be estimated indirectly. The first group of parameters includes variables such as survival probabilities and capital ratios. The second group of variables includes those that are estimated through a calibration procedure. The calibration procedure usually consists of adjusting the parameter values until one or more model outputs are sufficiently close to their observed real-world values.

3.1. Exogenous parameters

Each period in the model corresponds to 5 years in real life. Households are assumed to begin their economic life at age 20 $(j = 1)$ and face a life expectancy of 100 years, so the life cycle in the model covers $JJ = 16$ periods. The mandatory retirement age is defined as 65, which in the model means that the retirement age corresponds to $JR = 10$ periods, so households spend the last 7 periods as retirees and receive a pension.

Since we are considering a period to correspond to 5 years, some annual rates must be converted. Considering the case of the population growth rate, assuming an annual growth rate of 1 percent, the conversion to a 5-year compound annual growth rate would be equal to $n_p = 1.01^5 - 1 \sim 0.05$.

The capital-to-product ratio and the labor-to-income ratio were obtained from PWT 10.01, Penn World Table. The age-specific productivity profile was calculated using the National Survey of Occupation and Employment (ENOE) for the second quarter of 2021.

Total government spending as a fraction of GDP is obtained from the World Bank and corresponds to government final consumption expenditure[9], while the public debt-GDP ratio was obtained from the ECLAC database.

For the calibration of the model we consider 2021 as the base year, so that all exogenous parameters correspond to this year.

[8]Actual Pension Amount.
[9]https://data.worldbank.org/indicator/NE.CON.GOVT.ZS

3.2. Calibrated parameters

The parameters to be calibrated corresponding to production were the level of technology Ω and the depreciation rate δ . At the authors' suggestion, the wage rate is normalized to one, $w = 1$. The parameter Ω It was numerically calibrated until the wage rate values closest to one were obtained. Meanwhile, The depreciation rate did not need to be calibrated because the rate for all three countries is presented in PWT 10.01, Penn World Table.

The parameter ν It represents the trade-off of individual preferences regarding consumption and leisure. The larger the value of ν , It is more attractive for households to consume goods and services that are paid for in the market than to consume leisure time. The parameter ν , Therefore, it has a significant influence on the number of hours a household works in the market. It adjusts ν to a target of an average ratio of work time to total allocated time that represents approximately 33 percent. This value is calculated for each country assuming a maximum weekly work time allocation of 110 hours, as well as 50 work weeks per week. This annual average of hours worked per employee, which is available in PWT 10.01, Penn World Table for the countries under study, is then compared.

The following parameters to be calibrated correspond to the formation process and variance of the logarithm of wage income over the household life cycle. Empirical studies indicate that around age 25, the variance of income is 0.3 and tends to increase almost linearly to a value of 0.9 by age 60. In the model presented here, the variance of the logarithm of labor earnings is determined by two components: through exogenous processes that affect labor productivity in an idiosyncratic way θ and η_j , as well as by individual decisions about how many hours of work are offered in the market. We have information about the structure of the labor productivity process and how this might influence the variance of the logarithm of labor earnings. The logarithm of an individual's labor earnings is defined as

3.3. Tax system and pension system

The tax and pension systems still need to be defined. The government has four tax schemes to define in order to balance its budget:

1. Define exogenously the value of τ_t^w and τ_t^r , calculate the value of τ_t^c .
2. Define exogenously the value of τ_t^c , calculate the value of τ_t^w and τ_t^r
3. Define exogenously the value of τ_t^c and τ_t^r , calculate the value of τ_t^w .
4. Define exogenously the value of τ_t^c and τ_t^w , calculate the value of τ_t^r .

For the model executions, scheme 4 was defined, meaning we exogenously assign a value for the consumption and labor income tax rates, and the model calculates the capital tax rate. The calculations for the effective consumption and income tax rates performed by the CIEP were used.

With respect to the pension system, we have to define the replacement rate κ . The observed replacement rate value for the three countries was obtained from OECD-Founded Pension Indicators-Contributions.

The value of the intertemporal discount factor β was the same as that used by the authors.

For the initial equilibrium, we consider the pension system to be regressive, meaning the progressivity factor $\lambda = 0$.

The mortality probability rates were estimated using the population pyramids by age cohort from the 2020 Population and Housing Census.

3.4. Summary of Exogenous (E), Calibrated (C), and Target (T) Parameters

The following table presents the model parameters. They are classified according to parameters that are obtained directly from the data (exogenous parameters) and those that are calibrated. The calibration process is briefly described. For more detail, see the Model Parameterization and Calibration section.

The model outputs that are defined as calibration targets are also shown. For the case of these targets, as well as the exogenous parameters, the data source used in the present analysis is indicated.

Parameter	Description	E	C	T	Description
TT	Number of transition periods. Each period is equivalent to 5 real-life years.	X			Defined by numerical criterion
JJ	Number of years a household lives. Households begin their economic life at 20 years old ($j = 1$). They live until 100 years old ($JJ = 16$).	X			
JR	Mandatory retirement age. Households retire at 65 years old ($j_r = 10$)	X			
γ	Coefficient of relative risk aversion (reciprocal of the intertemporal substitution elasticity)		X		The parameter was calibrated until the outputs closest to the observed values of the Consumption and Investment ratios with respect to GDP were obtained.
ν	Leisure preference intensity parameter.	X			Consulted PWT 10.01, Penn World Table
β	Time discount factor.		X		Calibrated by Fehr and Kindermann (2018).
σ_θ^2	Variance of the fixed effect θ on productivity.		X		Calibrated by Fehr and Kindermann (2018).
σ_ε^2	Variance of the autoregressive component η .	X			Calibrated by Fehr and Kindermann (2018).
α	Capital elasticity in the production function. Corresponds to the capital-output ratio.	X			Consulted PWT 10.01, Penn World Table.

Parameter	Description	E	C	T	Description
δ	Capital depreciation rate.	X			Consulted PWT 10.01, Penn World Table
Ω	Technology level.				Numerically calibrated to adjust the wage rate to $w_t = 1$.
n_p	Population growth rate.	X			Consulted OECD, Fertility rates
gy	Public expenditure as a percentage of GDP.	X			Consulted World Bank, General government final consumption expenditure (% of GDP)
by	Public debt as a percentage of GDP.	X			CEPAL database
κ	Pension system replacement rate.	X			Consulted OECD-Founded Pension Indicators-Contributions
ψ_j	Survival rates by age cohort.	X			Calculated using the population pyramids from the 2020 Population Census
e_j	Labor income efficiency profile by age cohort.	X			ENOE Q2 2021
τ_t^c	Consumption tax rate.	X			Effective rate calculated by CIEP
τ_t^w	Labor income tax rate.	X			Effective rate calculated by CIEP
τ_t^r	Capital income tax rate.		X		Consulted OECD Tax Database
τ_t^p	Pension system payroll contribution rate.		X		Consulted OECD-Founded Pension Indicators-Contributions
$\tau_{j,t}^{impl}$	Implicit tax rate of the pension system payroll contribution.		X		
λ	Pension system progressivity factor.	X			
PEN/GDP	Pension payment as a percentage of GDP.		X		Consulted OECD-Pensions at Glance-Public expenditure on pensions
C/GDP.	Private consumption as a percentage of GDP.		X		Consulted PWT 10.01, Penn World Table
I/GDP	Investment as a percentage of GDP.		X		Consulted PWT 10.01, Penn World Table

3.5. Model parameters

The following table summarizes the model parameter values:

Description	Parameter	México
Función de Utilidad		
Relative risk aversion coefficient (reciprocal of the intertemporal elasticity of substitution)	γ	0.18
Parameter of leisure preference intensity.	ν	0.45
Time discount factor.	β	0.998
Production Function		
Elasticity of capital in the production function. It corresponds to the capital-output ratio.	α	0.619
Capital depreciation rate.	δ	3.8 %
Technology level.	Ω	1.65
Risk in labor productivity		
Autoregressive component of the shock in productivity.	ρ	0.98
Variance of the autoregressive component η	σ_ε^2	0.05
Government		
Public spending as a percentage of GDP.	gy	11.02 %
Public debt as a percentage of GDP.	by	52.3 %
Consumption tax rate.	τ_t^c	5.73 %
Income tax rate.	τ_t^w	12.73 %
Pension System		
Pension system replacement rate.	κ	64.3%
Progressivity Factor of the pension system.	λ	0.0

Description	Parameter	México
Demography		
Population growth rate	n_p	1.8 %

3.6. Initial Equilibrium

	Model	Observed
Mercado de Bienes (% PIB)		
• Private Consumption	69.04	70.2
• Public Spending	11.02	11.02
• Investment	19.94	18.39
Tasas de impuestos (en %)[10]		
• Consumption	5.73	16.0
• Income	12.73	1.92 - 35.0
• Capital	14.58	12.13[11]
Tax revenue (% of GDP)		
• Consumption	3.96	4.07
• Income	4.85	3.91[12]
• Capital	8.14	5.33
Pension System		
• Replacement rate	64.3	64.3
• Pension payments (% of GDP)	2.44	18.2
Demography		
• Population growth rate	1.82	1.82

4. Increased Progressivity of the Pension System

This section will detail the macroeconomic, welfare and efficiency effects, as well as the fiscal sustainability, of increasing the progressivity of the pension system.

We start from the initial equilibrium obtained in the previous section, which was defined as $t = 2021$. We progressively change the value of λ starting from $t = 2022$ and calculate the transition path towards the new equilibrium state $t = \infty$.

We are interested in understanding the effects on macroeconomic aggregates, as well as on welfare and efficiency, between both equilibrium states.

4.1. Calculations of effects on welfare and efficiency

In order to account for changes in the well-being of the agents, we will use the so-called **Hicksian Equivalence**[13].

Given that the household utility function is homogeneous [3], we have that the following equality holds.

$$u\big[(1+\phi)c_j, (1+\phi)\ell_j\big] = (1+\phi)u\big[c_j, \ell_j\big] \tag{38}$$

If consumption and leisure were simultaneously increased by a factor of $1+\phi$ at any age, lifetime utility would increase by the same factor. If we assume that an individual in state z_j has utility $V_{2021}(z_j)$ in the initial equilibrium and $V_t(z_j)$ in the second long-run equilibrium after the policy, $t > 2021$.

The compensation variation between the reform scenario and the baseline scenario for the individual with status z_j would be

$$\phi = \frac{V_t(z_j)}{V_{2021}(z_j)} - 1 \tag{39}$$

where ϕ indicates the percentage change in both consumption and leisure that the individual in state z_j would require in the initial state to be at least as well off as after the policy reform. If $\phi > 0$, the reform improves the well-being of this individual, and vice versa.

[10]Effective Rates vs Nominal Rate
[11]Effective Rate
[12]Income taxes applied to wages and mixed income
[13]Compensation Variation

The solid line in the figure below shows a possible cohort-specific consequence on well-being that individuals may experience as a result of a policy[14]. The policy redistributes wealth from current cohorts to future cohorts[15]

To isolate the effects solely attributable to the efficiency of the reform, [4] pioneered the introduction of a new agent into the model: the Lump-Sum Redistribution Authority (LSRA). This agent performs a hypothetical task in an independent simulation, redistributing and compensating for the benefits or losses generated by the policy. First, this agent makes transfers or imposes taxes on all economically active generations in the year immediately preceding the policy's implementation, aiming to ensure they are as well off as they were in the initial equilibrium state after the policy. Therefore, its compensating variation is equal to 0. Subsequently, as a consequence of this redistributive operation, the LSRA may have incurred debt or accumulated assets. The LSRA redistributes this debt or assets to future generations in such a way that everyone receives the same compensating variation. This variation can be interpreted as a measure of efficiency. [3] If this variation is positive, the reform is considered to be Pareto improved after compensation.

4.2. Effects of increased progressivity

We are interested in calculating the effects of increasing the progressivity of the pension system and calculating the path to the second equilibrium state for the values of $\lambda \in \{0.1, 0.2, \ldots, 1.0\}$. This intervention has two latent effects on the system. The first: Under the earning points system, a portion of workers' pension contributions is recognized as implicit savings. Under the system where the pension is independent of income history ($\lambda = 1$), the contribution is considered an implicit tax. Thus, the transition to a pension system with a flat pension increases distortions in the labor supply and decreases welfare. The second: The pension system independent of income history ($\lambda = 1$) also functions as insurance against labor market risks, which tends to improve welfare.

4.2.1. Macroeconomic effects

4.2.2. Well-being and Efficiency

We use Hicksian variation as a measure of the effects on well-being for different cohorts. We would like to answer:

- Are retirees affected or benefited? How are they affected by increases in consumption taxes? Are those in the formal or informal sectors more affected or harmed?
- Are current working-age generations affected or benefited?
- Here the intra-generational redistribution from rich towards poor households induced by the progressive pension formula becomes most obvious.
- Are the new generations affected or benefited?
- What about the overall measure of the LSRA? That is, the welfare effects after compensation payments.

The evolution of welfare after compensation is depicted in the right part of Table 5. We find that the reform induces losses for any future generation of 0.46% of initial resources.

The introduction of flat pensions comes along with two major efficiency consequences:
- on the one hand, insurance provision against labor market risk causes efficiency to rise while,
- on the other hand, increasing labor market distortions reduce it.

We find in this reform scenario that the latter outweighs the former and therefore the introduction of completely flat pension benefits is Pareto inferior.

Table 1: Welfare effects of flat pensions (base model version).

Año	**2022**	**2030**	**2040**	**2050**	**2060**	**2070**	**2080**	**2090**	∞
Agregados Macroeconómicos									
PIB	−0.32	−2.15	−3.56	−4.34	−5.15	−5.17	−5.19	−5.19	−5.19
Trabajo	−0.82	−1.35	−1.29	−1.26	−1.25	−1.24	−1.23	−1.23	−1.23
Capital	−0.0	−2.63	−4.93	−6.18	−7.05	−7.3	−7.53	−7.51	−7.51
Precios									
Salario	0.51	−0.81	−2.3	−3.11	−3.58	−4.01	−4.03	−4.03	−4.03
Tasa de interés	−0.39	0.63	1.83	2.49	2.87	3.09	3.22	3.21	3.21
Impuesto al capital	0.36	4.54	8.18	10.21	11.38	12.05	12.08	12.07	12.07
Sistema de Pensiones									
Gasto en Pensiones[16]	2.44	2.87	2.84	2.84	2.84	2.84	2.84	2.84	2.84
Tasa de Contribución	−0.0	17.62	16.35	16.35	16.35	16.35	16.35	16.35	16.35

[14] A representative agent is considered for each cohort.

[15] The axis x of the graph indicates the birth year of the cohorts. Since households begin their working lives at age 20, following the example of [3], the last cohort to enter the labor market was born in 1998, as the first equilibrium state corresponds to 2008. When referring to future generations, this means generations born after 2009.

[16] en % de PIB

Referencing 1 in-text using its label.

Table 2: Welfare effects of flat pensions (base model version).

Birth Year	**Years in 2021**	**Sin LSRA**		**Con LSRA**
		Por tipo de Empleo		
Retired		**Formal**	**Informal**	
1930	91	1.19	0.53	0.0
1956	65	3.93	2.47	0.0
Workers		**Formal**	**Informal**	
1960	61	−7.87	2.01	0.0
1970	51	−4.43	0.80	0.0
1980	41	−0.65	0.52	0.0
1990	31	0.47	0.28	0.0
2000	21	−0.73	−0.73	0.0
Future Generations				
2012	19			−0.13
2020	1			−0.13
2040	-			−0.13
2060	-			−0.13

Table 3: Aggregate efficiency effects of alternative progressivity levels.

λ	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**	**0.7**	**0.8**	**0.9**
Macroeconomic Aggregates									
GDP	−0.56	−1.12	−1.69	−2.26	−2.82	−3.39	−3.96	−4.51	−5.03
Work	−0.11	−0.24	−0.36	−0.48	−0.6	−0.73	−0.85	−0.98	−1.1
Capital	−0.82	−1.66	−2.5	−3.33	−4.16	−4.99	−5.82	−6.63	−7.37
Prices									
Wage	−0.44	−0.88	−1.33	−1.78	−2.23	−2.68	−3.13	−3.57	−3.97
Interest rate	0.34	0.69	1.05	1.41	1.77	2.14	2.5	2.86	3.2
Capital tax	1.29	2.61	3.95	5.3	6.67	8.06	9.44	10.83	12.12
Sistema de Pensiones									
Pensions expenditure[17]	2.48	2.52	2.56	2.6	2.64	2.68	2.72	2.76	2.8
Contribution Rate	1.57	3.16	4.76	6.37	8.0	9.64	11.3	12.97	14.66
Eficiencia Agregada									
With LSRA	−0.01	−0.02	−0.03	−0.041	−0.053	−0.065	−0.079	−0.094	−0.111

[17] en % de PIB

Table 4: Welfare effects of flat pensions (base model version) and Complete Formal Labor Market.

Año	**2022**	**2030**	**2040**	**2050**	**2060**	**2070**	**2080**	**2090**	∞
Agregados Macroeconómicos									
GDP	−0.25	0.84	0.38	−0.0	−0.22	−0.34	−0.41	−0.44	−0.47
Work	−0.65	4.02	3.69	3.71	3.71	3.72	3.72	3.72	3.72
Capital	−0.0	−1.06	−1.61	−2.22	−2.56	−2.75	−2.86	−2.92	−2.96
Precios									
Wage	0.4	−3.06	−3.2	−3.58	−3.79	−3.91	−3.98	−4.02	−4.04
Interest rate	−0.31	2.44	2.56	2.87	3.05	3.15	3.2	3.23	3.25
Capital tax	0.11	−0.59	0.3	1.23	1.75	2.05	2.22	2.31	2.37
Sistema de Pensiones									
Pensions expenditure[18]	2.42	2.71	2.97	2.97	2.97	2.97	2.97	2.97	2.97
Contribution Rate	−4.16	−2.05	7.6	7.6	7.6	7.6	7.6	7.6	7.6
Eficiencia Agregada									
With LSRA									1.05

5. Conclusions

The increase in the progressivity of the pension system has positive effects on aggregate efficiency only if it is accompanied by an increase in the formality rate.

This is because the positive effect of a progressive pension, functioning as a coverage mechanism against random shocks, expands **vis a vis** the negative effect that distorts the labor market.

The combination between fiscal sustainability and efficiency increase is dominated by the level of the formality rate rather than the level of the system's progressivity.

Next steps:

- **Pensions**: Explore schemes for contributory and non-contributory pension systems.
- **Labor Market**: definition of formality, adjusting the production function. 28% of the employed population is outside the current definition.
- **Care**: Incorporation of women into the labor market and financing of the care system. **Unpaid care work is equivalent to 24.5% of GDP (70% performed by women)** (INEGI).
- **Health**: Incorporate categorization of health status (good/bad health). More than 50% of the 50+ population has diabetes or hypertension or both. 60% of years lost due to disability (DALYs) from NCDs (2021) (IHME).
- Model the impact of changes in fiscal rules.

A. Appendix A

A.1. Solución al problema de los hogares

Para encontrar la solución al problema de los hogares se necesita discretizar el espacio de estados z . Se tiene que calcular tres conjuntos de nodos

$$\hat{\mathcal{A}} = \{a^1, ..., a^{n_A}\}, \hat{\mathcal{P}} = \{ep^1, ..., ep^{n_P}\}, \hat{\mathcal{E}} = \{\eta^1, ..., \eta^{n_E}\} \tag{A.1}$$

Se usa el método de [5] para obtener una aproximación de la distribución de η , el cual sigue un proceso AR(1), mediante una Cadena de Markov discreta.

Para cada uno de los valores discretizados de z_j se calcula la decisión óptima de los hogares a partir del problema de optimización (función de política) mediante el algoritmo de iteración de la función de política el cual utiliza una interpolación spline multidimensional [6] del nivel de ahorro y earning points de los hogares así como el método de Newton para encontrar las raíces de la condición de primer orden.

A.2. Algoritmo para el equilibrio macroeconómico del cálculo del equilibrio inicial y transición

Las series de tiempo de precios de los factores así como los valores de las variables de política de la transición del estado de equilibrio inicial al siguiente se obtienen mediante el algoritmo iterativo Gauss-Seidel [4].

[18] en % de PIB

Se fijan las condiciones iniciales de las variables de stock K_1 , BQ_1 , B_1 , capital, herencias y deuda respectivamente. Se asignan iguales a los valores del equilibrio inicial K_0 , BQ_0 B_0 .

El valor de dichos stocks es calibrado a lo largo de la transición mediante un parámetro de velocidad de ajuste **damp factor**.

El pseudocódigo del programa de la transición es el siguiente:

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